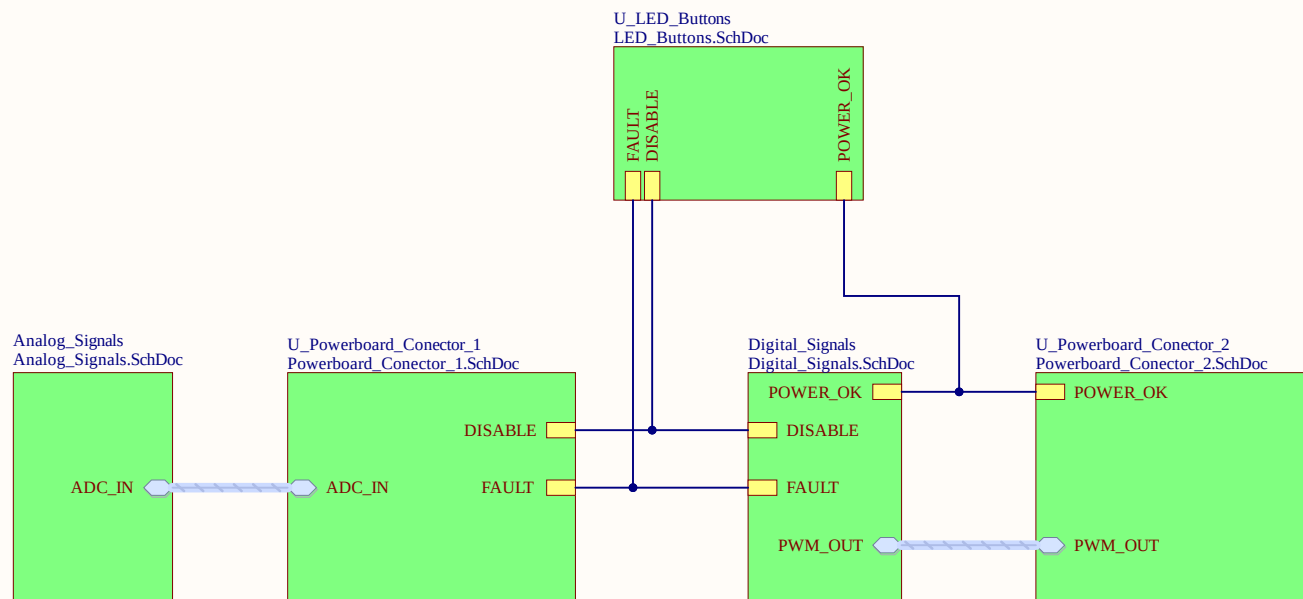
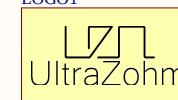
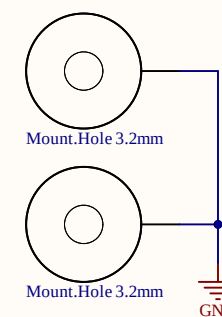




LOGO1



UZ Logo



Title Interface_Board_Top.SchDoc

Revision: 1v00

Design Engineer: Eugen Romaschenko

Project: MDK_UltraZohm_Adapter.PrjPcb

UltraZohmwww.ultrazohm.com

Date: 09.02.2021

Sheet 1 of 6



A

B

C

D

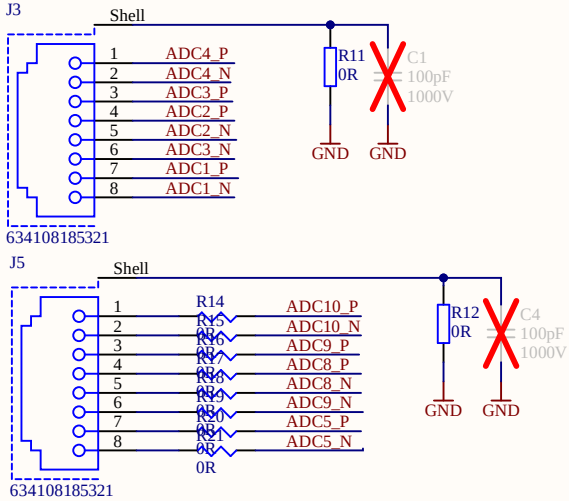
A

B

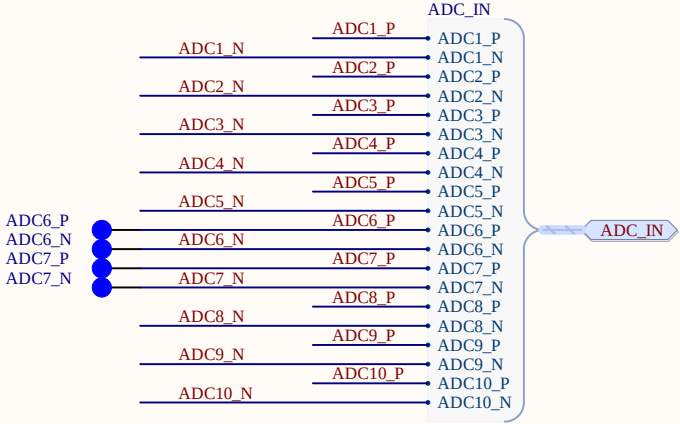
C

D

ADCs 1-5 and 8-10 are used



OnSemi Output Range = 0 to 3.3V to GND
The signals are not fully differential but referenced to GND



A

A

B

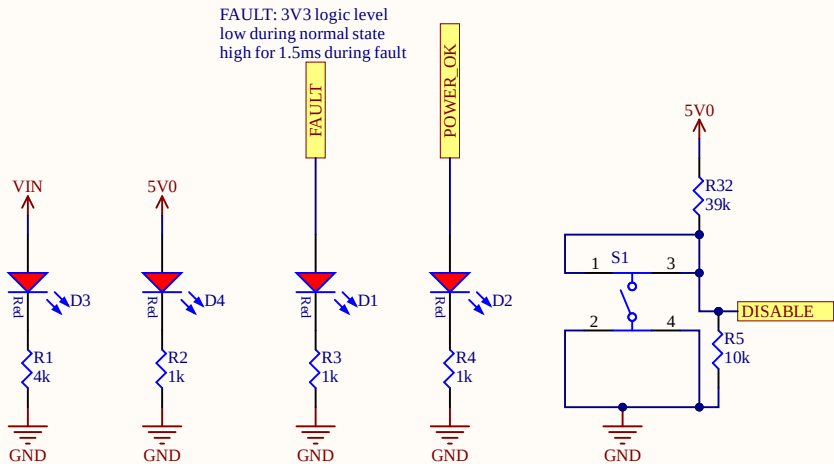
B

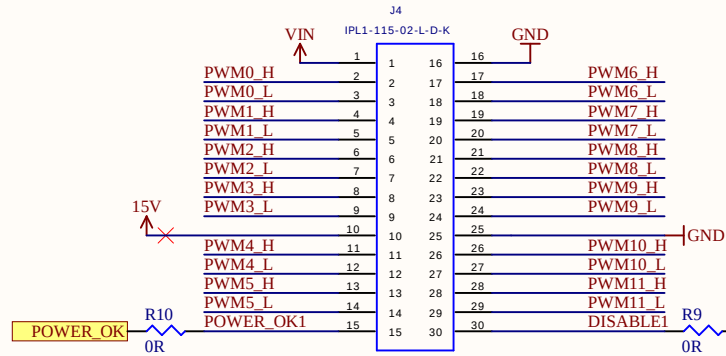
C

C

D

D





PWM Voltage Levels for NFAQ1060L36T Driver IC

Maximum PWM Input Voltage = VDD = 15V

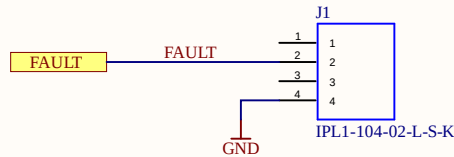
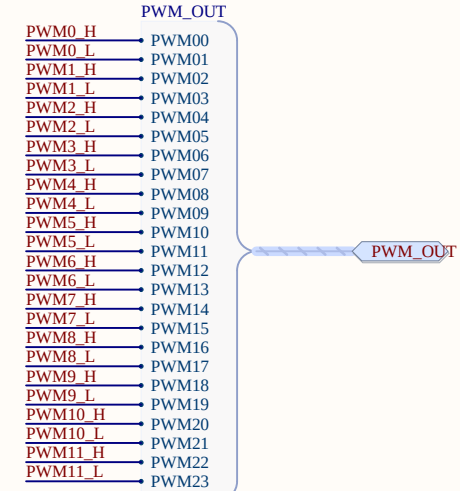
Recommended PWM Input Voltage ON Voltage min = 3V max = 5V

Recommended PWM Input Voltage OFF Voltage min = 0V max = 0.3V

<https://www.onsemi.com/pub/Collateral/NFAQ1060L36T-D.PDF>

The pin out on J4 does not match J1 on the UZ voltage adapter card on purpose. On the MDK these are not differential signals, but the high and low side gate signals. Since this is the largest connector that would fit on this adapter board, we opted for using the "small one" with the following modifications:

- For up to 12 gate signals, a pre-fitted standard samtec cable can be used, making pins J4-1 to J4-15 available, i.e. PWM0 to PWM5.
- If more than 12 gate signals are needed, the second row (starting at pin 17) needs to be connected to second connector on the voltage adapter card. In this case the samtec connector on UZ side has to be modified.



A

A

B

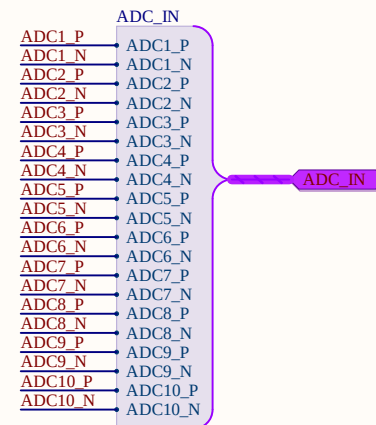
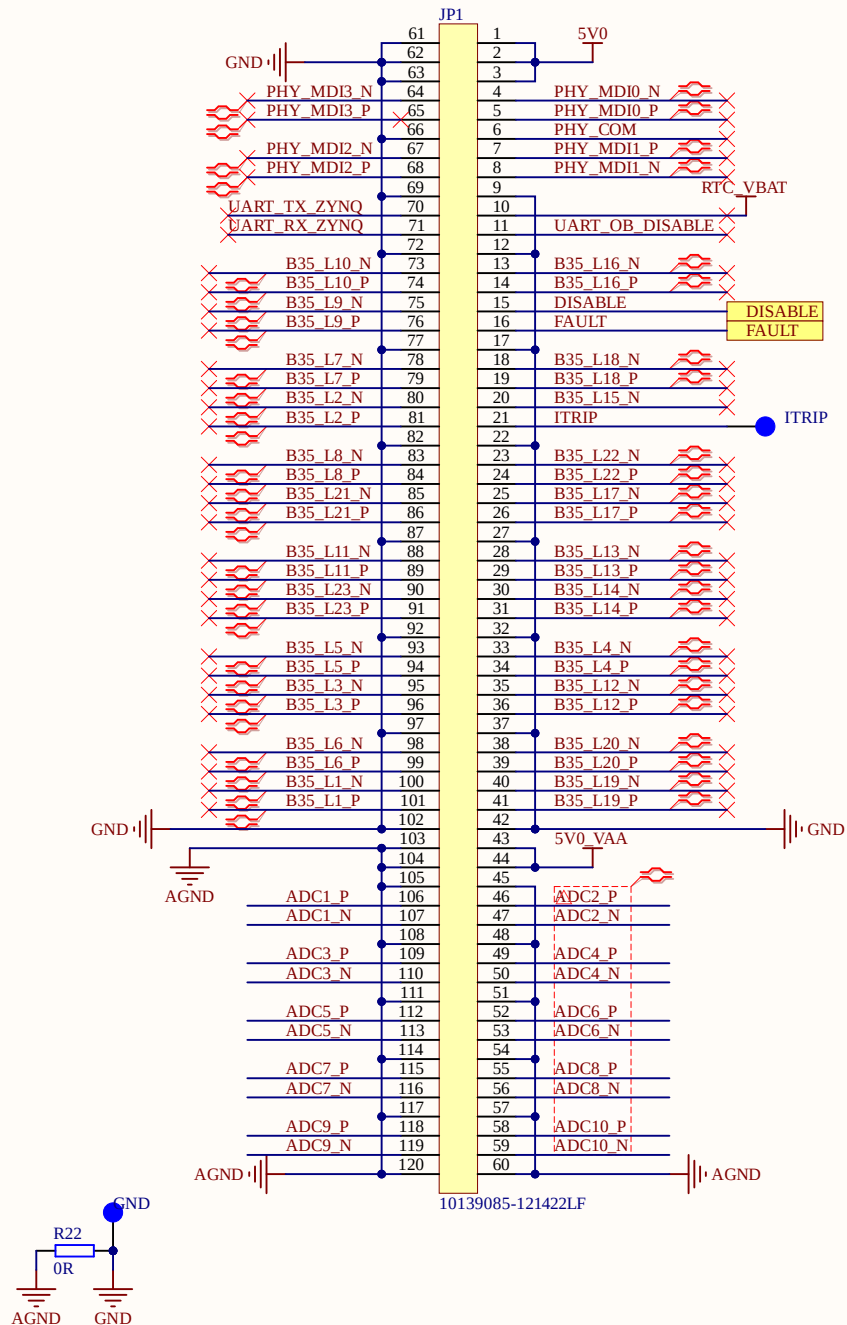
B

C

C

D

D



ADC Voltage Range = 0 to 3.3V
FAULT Voltage Level = 3.3V

Title Powerboard_Conector_1.SchDoc

Revision: 1v00

Design Engineer: Eugen Romaschenko

Project: MDK_UltraZohm_Adapter.PrjPcb

UltraZohm

www.ultrazohm.com

Date: 09.02.2021

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